

Covering Space

1.1 Basic Definitions

Definition 1.1.1 ► Covering Space

Let $p : \bar{X} \rightarrow X$ be a surjective map.

- We say an open subset V of X is **evenly covered** by p , if $p^{-1}(V)$ is a disjoint union of open subsets of $\bar{X}$:

$$p^{-1}(V) = \coprod_i U_i$$

where, each U_i is mapped homeomorphically onto V by p .

- In this case, we shall refer to each U_i as a **sheet** for p over V .
- If the space X can be covered by open subsets which are evenly covered by p , then we say that p is a **covering projection**; the space $\bar{X}$ is called a covering space of X .

Theorem 1.1.2

Let $p : \bar{X} \rightarrow X$ be a continuous map, where X is locally path connected.

1. The map p is a covering space iff for each component $\bar{C}$ of $\bar{X}$, the restriction map $p : p^{-1}(C) \rightarrow C$ is a covering projection.
2. If p is a covering projection then for each component $\bar{C}$ of $\bar{X}$, the map $p : \bar{C} \rightarrow p(\bar{C})$ is a covering projection and $p(\bar{C})$ is a component of X .

Remark.

Thus, we always assume that a covering space is locally path connected and connected. For the following content, unless otherwise specified, we will all adopt this assumption.

1.2 Lifting Properties

Theorem 1.2.1 ► The uniqueness of map's lifting

Let $p : \bar{X} \rightarrow X$, be a covering projection, Y be any connected space and $f : Y \rightarrow X$ be any map. Let $\tilde{f}_1, \tilde{f}_2 : Y \rightarrow \bar{X}$ be any two lifts of f , such that, for some point $y \in Y$, $\tilde{f}_1(y) = \tilde{f}_2(y)$. Then $\tilde{f}_1 = \tilde{f}_2$.

Theorem 1.2.2 ► Path lifting property

Let $p : \bar{X} \rightarrow X$ be a covering projection. Then given a path $\omega : \mathbb{I} \rightarrow X$ and a point $\bar{x} \in \bar{X}$ such that, $p(\bar{x}) = \omega(0)$, there exists a path $\bar{\omega} : \mathbb{I} \rightarrow \bar{X}$ such that, $p \circ \bar{\omega} = \omega$ and $\bar{\omega}(0) = \bar{x}$.

Theorem 1.2.3 ► Homotopy Lifting Property (HLP)

Every covering projection $p : \bar{X} \rightarrow X$ is a fibration. That is for each space Y , map $\tilde{f} : Y \rightarrow \bar{X}$, and for each homotopy $F : Y \times [0, 1] \rightarrow X$ such that $F(y, 0) = p(\tilde{f}(y))$, there exists a lifting $\tilde{F} : Y \times [0, 1] \rightarrow \bar{X}$, satisfying

1. $\tilde{F}(y, 0) = \tilde{f}(y)$
2. $p \circ \tilde{F} = F$.

1.3 Relation with the Fundamenatal Group